

CONVERSATIONAL INVERSE DESIGN OF CELLULAR STRUCTURES IN IMPLICIT FIELD SPACE

Yunsheng Wang¹, Shaoliang Yang¹, Jun Wang¹

¹Santa Clara University, Santa Clara, CA

ABSTRACT

In this paper, we introduce a conversational inverse design framework that lets a user describe the behavior they want in natural language and receive multiple candidate unit-cell geometries represented directly in implicit field space. The system converts each prompt into explicit targets and constraints, retrieves and ranks a diverse set of implicit-field candidates that aim to satisfy those requirements, and verifies them with a surrogate model to maintain physical fidelity. To keep the pipeline controllable and reproducible, we also release an implicit-field dataset of cellular structures built from analytic families and parameterized transformations, with metadata that enables principled control over periodicity. We evaluate the method using target-matching error, success rate under strict tolerances, diversity among valid solutions, and computational cost, and we benchmark a strong retrieval baseline under the same verification protocol. The results suggest that conversational conditioning can be a practical interface for exploring multimodal cellular structure design while still providing performance guarantees.

1. INTRODUCTION

Inverse design reverses the usual engineering loop: we begin from desired behavior and search backward for a geometry that realizes it. This is attractive whenever evaluation is expensive—for example, when each candidate requires a numerical solve, a homogenization routine, or high-fidelity simulation. Classical approaches such as topology optimization provide a principled route to high-performance structures, but they are typically iterative, solver-intensive, and often produce a single design outcome for a given formulation [1, 2]. In many modern design settings, engineers instead want rapid exploration: a short list of viable options, each meeting explicit targets and constraints.

Recent progress in machine learning has made fast inverse mappings practical across multiple scientific domains. Data-driven models have accelerated inverse design in nanophotonics [3], enabled learning-based modeling and control in complex dynamical and fluid systems [4], and supported materials and molecular discovery pipelines that connect objectives to latent

or learned descriptors [5–7]. These successes share a common promise: once trained, a model can propose candidates at a small fraction of the cost of repeatedly calling a high-fidelity solver.

Cellular structures are a compelling but challenging target for inverse design. A unit cell’s topology and morphology govern effective stiffness, density, anisotropy, and failure mechanisms, yet the inverse map is inherently one-to-many: many distinct microstructures can achieve similar effective responses. Recent work in mechanical metamaterials has therefore moved beyond single-solution inverse maps toward conditional generative models that explicitly represent multi-modality and can return diverse candidates under the same property targets [8]. However, two gaps remain for practical engineering use: (i) controllable geometric representations that remain stable under topology changes, and (ii) reliable verification so that generated candidates satisfy quantitative requirements rather than merely looking plausible.

Implicit geometry, especially signed distance fields (SDFs), offers a robust design domain for cellular structures. SDFs represent geometry as a continuous volumetric field whose zero level set defines the surface, naturally handling complex topology without brittle meshing pipelines. Learning-based SDF representations have shown high-fidelity shape modeling and interpolation [9], and implicit neural representations with periodic activations further improve the ability to capture high-frequency structure and derivatives that matter for geometry processing [10]. For periodic cellular unit cells, implicit fields also make it straightforward to compute density proxies, extract surfaces at arbitrary resolution, and define boundary-consistency measures that serve as practical periodicity proxies.

A second practical gap is interface. Engineers think in intent—“lighter but stiff,” “more isotropic,” “keep pores open but connected,” “give me a few options”—not in latent vectors. Language-driven design tools have advanced rapidly, with work on text-guided CAD and text-to-3D asset generation demonstrating that natural language can be a viable front-end for geometric synthesis [11, 12]. Yet engineering design requires a different contract with the user: the output must satisfy explicit targets and constraints, and the system should provide verification signals

that support trust and reproducibility.

In this paper, we propose a conversational inverse design framework that couples a language interface with an implicit-field design space and a verification-first loop. A user provides a natural-language request; the system parses it into structured targets and constraints, proposes multiple candidate unit cells in implicit space, and verifies and ranks candidates via fast property prediction validated against homogenization when available [13, 14]. To keep the pipeline controllable and reproducible, we also release a large implicit-field dataset of cellular structures built from analytic families and parameterized transformations, complementing existing catalogs of equation-based lattices and homogenized property data [15]. Overall, our goal is to make conversational conditioning a practical interface for exploring multimodal cellular-structure design while maintaining quantitative guarantees.

Overall, the contributions of this paper are:

1. We introduce a large-scale implicit-field dataset of periodic cellular structures generated from analytic families with parameterized transformations. Each sample includes ground-truth metadata that enables controlled studies of periodicity and morphology.
2. We present a conversational inverse design pipeline that translates natural-language intent into structured targets and constraints, and proposes multiple candidate unit cells via retrieval in implicit field space to reflect the one-to-many nature of the problem.
3. We define a verification-first evaluation protocol that reports target-matching error, success rate under strict tolerances, diversity among valid solutions, and computational cost, and we benchmark against a strong retrieval baseline with constraint filtering and diversity selection.

2. RELATED WORKS

Our work sits at the intersection of (i) inverse design for materials and architected structures, (ii) implicit geometry representations such as signed distance fields, and (iii) human-in-the-loop interfaces that make high-dimensional design spaces easier to explore. Below we review the threads most relevant to conversational inverse design in implicit field space.

2.1. Inverse Design for Structures and Metamaterials

Inverse design asks for a geometry or a set of design variables that achieves a target response, often under tight constraints. In classical structural optimization, topology optimization provides a principled route to high-performance layouts, but it is typically iterative and expensive when each evaluation requires a PDE solve [1, 2]. For microstructured materials, the challenge is amplified by multiscale effects: the unit cell geometry determines effective properties, while the macro-scale application determines the desired behavior.

Data-driven inverse design has become a practical alternative when paired with efficient forward models. In nanophotonics, deep networks have been used to map target spectra to device geometries, accelerating design exploration [3]. Similar ideas

appear across scientific inverse problems and materials discovery [4–6]. In mechanical metamaterials, recent work has emphasized that the inverse map is inherently one-to-many: many distinct geometries can realize the same effective targets. Generative approaches explicitly model this multi-modality, producing diverse candidates conditioned on properties [8]. Diffusion and other modern generative models have also been explored for property-conditioned lattice generation, improving sample quality and controllability [9]. These methods motivate our focus on producing *multiple* valid unit cells per prompt rather than a single “best” design.

2.2. Homogenization and Property Evaluation for Cellular Structures

Most inverse design pipelines for periodic cellular materials depend on a reliable way to evaluate effective properties. Homogenization methods provide the standard bridge from micro-geometry to effective tensors and moduli [13]. For common cellular families such as TPMS-like lattices, property computation is often implemented with voxel-based or FE-based periodic boundary conditions, enabling large-scale data generation and benchmarking [8]. As datasets and catalogs have grown, researchers have begun releasing curated libraries of implicit or equation-based unit cells together with homogenized property data, supporting systematic comparisons across model classes [15].

Our paper builds on this tradition but emphasizes two gaps that matter in practice: (1) controllable datasets that let us isolate geometric factors (not just sample more shapes), and (2) verification mechanisms that keep selection tied to physically meaningful targets, instead of relying on plausibility alone.

2.3. Implicit Geometry and Signed Distance Fields

Implicit representations encode geometry as a continuous field, typically with the surface defined as a level set. Signed distance fields (SDFs) are especially useful because they represent complex topology without explicit meshing and provide a smooth signal for learning and optimization. In graphics and vision, DeepSDF showed that neural networks can learn continuous SDFs for shape representation and reconstruction [9]. Related work on implicit decoders (e.g., IM-NET) demonstrated that implicit fields can also serve as a strong generative representation [10]. Meanwhile, implicit neural representations with periodic activations improved fidelity for high-frequency detail and derivatives, which are often important for geometry and physics [10]. These advances have led to a broad ecosystem of implicit-shape generation methods and surveys [11].

In engineering settings, implicit fields are attractive for cellular structures because periodicity, blending, and morphological control can be implemented as field operations, and surfaces can be extracted at arbitrary resolution when needed. This makes implicit space a natural design domain for both learned models and downstream evaluation.

2.4. Language Interfaces and Human-in-the-Loop Generative Design

Natural language interfaces for CAD have been explored for decades in various forms, but only recently have large language models made free-form intent specification realistic at scale [11?]. In parallel, text-guided 3D generation has advanced rapidly in computer graphics, with methods that generate 3D assets from prompts using implicit representations or hybrid pipelines [12? ?]. These systems are compelling for visual content creation, but they typically do not enforce engineering constraints or guarantee that generated shapes meet quantitative physical targets.

Our work draws inspiration from these language-driven interfaces but targets a different contract with the user: not only “make something that matches my description,” but “return multiple candidates that satisfy explicit property targets and constraints, with verification.”

3. METHODOLOGY

We propose a conversational inverse design system for periodic cellular unit cells in implicit field space. Given a natural-language request, the system (i) parses intent into structured targets and constraints, (ii) proposes multiple candidate SDF unit cells from a controllable dataset, and (iii) verifies and ranks candidates using fast property prediction validated against homogenization when available.

3.1. Implicit-Field Dataset: OpenSS-SDF

3.1.1. Representation. Each unit cell is represented as an SDF volume $X \in \mathbb{R}^{1 \times D \times H \times W}$ on a fixed grid, where the surface is the 0-level set. We use a cubic domain $[-1, 1]^3$ with default resolution 32^3 .

3.1.2. Analytic function families. We generate SDFs from a library of analytic implicit functions. For example,

$$f_1(x, y, z) = \cos(2\pi x) + \cos(2\pi y) + \cos(2\pi z) - \text{ffscale}, \quad (1)$$

where ffscale shifts the level set and controls relative density. We include multiple trigonometric families and mixture families (Table 1) and optionally TPMS families (e.g., Gyroid, Schwarz D, Neovius). Each sample stores a discrete family label $y \in \{1, \dots, K\}$ and continuous parameters (e.g., ffscale).

3.1.3. Periodicalization transform. To control periodicity strength reproducibly, we apply a coordinate modulation. Let (xx, yy, zz) be grid coordinates. We compute

$$\text{PER} = xx + yy + zz, \quad \tilde{\text{PER}} = \frac{\text{PER} - \min(\text{PER})}{\max(\text{PER}) - \min(\text{PER})}, \quad (2)$$

then apply an exponent r and rescale the coordinates:

$$\hat{\text{PER}} = 0.5 + 0.5(1 - \tilde{\text{PER}})^r, \quad (\tilde{x}, \tilde{y}, \tilde{z}) = \left(\frac{xx}{\hat{\text{PER}}}, \frac{yy}{\hat{\text{PER}}}, \frac{zz}{\hat{\text{PER}}} \right), \quad (3)$$

where $r \sim \mathcal{U}(r_{\min}, r_{\max})$ controls deformation strength (larger r increases modulation and reduces periodic regularity).

3.1.4. Distance normalization. Given a function f and transformed coordinates $\mathbf{p} = (\tilde{x}, \tilde{y}, \tilde{z})$, we compute

$$X(\mathbf{p}) = \frac{f(\mathbf{p})}{\|\nabla f(\mathbf{p})\| + \epsilon}, \quad (4)$$

which improves scaling consistency across families.

3.1.5. Metadata and splits. Each sample stores $(X, y, \text{ffscale}, r)$ and summary statistics. In our experiments, we generate $N = 20,000$ samples and split them into train/val/test as 70%/15%/15%. The sampling ranges are $\text{ffscale} \sim \mathcal{U}(0.0, 0.5)$ and $r \sim \mathcal{U}(0.3, 0.9)$.

3.2. Conversational Interface: Prompt to Structured Request

3.2.1. Request schema. A user query q (natural language) is mapped to a structured request

$$\mathcal{R} = \{\mathbf{t}, \mathbf{c}, k\}, \quad (5)$$

where $\mathbf{t}$ are property targets (e.g., stiffness proxy, anisotropy score), $\mathbf{c}$ are constraints (e.g., density bounds, minimum periodicity), and k is the requested number of candidates.

3.2.2. Parsing model. We encode text into an embedding $e = \text{Enc}(q)$ and predict $(\hat{\mathbf{t}}, \hat{\mathbf{c}}, \hat{k}) = g(e)$. In this paper, we use a deterministic template-to-schema parser (rule-based) that maps qualitative descriptors (e.g., “lower density”, “more periodic”, “stiffer in x ”) to numeric targets/constraints in a fixed JSON schema.

3.2.3. Synthetic prompt generation. We build a prompt suite by templating over target ranges and qualitative descriptors (e.g. “more periodic”, “lower density”, “more isotropic”) and pair each prompt with ground-truth $(\mathbf{t}, \mathbf{c})$ derived from dataset metadata and property labels.

3.3. Candidate Proposal via Retrieval in Implicit Field Space

Given a structured request $\mathcal{R} = \{\mathbf{t}, \mathbf{c}, k\}$, our system proposes candidates by **retrieval and selection** from the implicit-field dataset. Specifically, we precompute a compact property vector for each dataset item (density proxy, periodicity proxy, and a stiffness label when available), then retrieve candidates whose properties are closest to the requested targets while satisfying constraints.

3.3.1. Constraint filtering. We first filter the dataset by hard constraints $\mathbf{c}$ (e.g., density bounds and minimum periodicity). If the feasible set is empty, we relax constraints in a fixed order (periodicity, then density) to return the closest feasible candidates and report the relaxation to the user.

3.3.2. Nearest-neighbor retrieval. Among feasible items, we compute a target-matching score in the verified property space,

$$d(X) = \|\hat{\mathbf{p}}_{\mathcal{T}}(X) - \mathbf{t}\|_2, \quad (6)$$

and take the top- M nearest neighbors (we use $M = 50$ in all experiments).

Function	Description
f1	$f(x, y, z) = \cos(2\pi x) + \cos(2\pi y) + \cos(2\pi z) - \text{ffscale}$.
f33	$\cos(2\pi(x + y - z)) + \cos(2\pi(x - y + z)) + \cos(2\pi(x - y - z)) - \text{ffscale}$.
f23I	$\cos(2\pi(x + y)) + \cos(2\pi(x - y)) + \cos(2\pi(x + z)) - \text{ffscale}$.
f23II	$\cos(2\pi(x + y)) + \cos(2\pi(x + z)) + \cos(2\pi(y + z)) - \text{ffscale}$.
f25IV	$\sin(2\pi(x - y)) + \sin(2\pi(x + z)) + \sin(2\pi(x - z)) + \sin(2\pi(y + z)) - \sin(2\pi(y - z)) - \text{ffscale}$.
f34I_f86I	Weighted combination of two trigonometric terms (full list in Appendix).

TABLE 1: Example analytic function families used to generate implicit fields.

3.3.3. Diversity selection. To reflect the one-to-many nature of inverse design, we select k candidates from the retrieved pool using a greedy diversity criterion. Starting from the best-scoring candidate, we iteratively add the candidate that maximizes the minimum distance to the already-selected set under ℓ_2 distance in SDF space (or in property space when only property vectors are available).

3.4. Property Verification and Ranking

3.4.1. Property vector. For each candidate X , we predict a property vector $\hat{\mathbf{p}}(X) \in \mathbb{R}^M$ including: (i) **relative density** $\rho(X)$ computed by voxel occupancy $\mathbb{I}(X < 0)$, (ii) **stiffness-related measures** (either homogenized or surrogate-predicted), (iii) **anisotropy score** $a(X)$ derived from the effective stiffness tensor (definition fixed across methods), and (iv) a **periodicity score** $s_{\text{per}}(X)$.

3.4.2. Periodicity score (proxy). We measure boundary consistency by comparing opposite faces of the SDF:

$$s_{\text{per}}(X) = 1 - \frac{1}{3} \left(\frac{\|X_{x=-1} - X_{x=+1}\|_1}{\|X_{x=-1}\|_1 + \|X_{x=+1}\|_1 + \epsilon} + \frac{\|X_{y=-1} - X_{y=+1}\|_1}{\|X_{y=-1}\|_1 + \|X_{y=+1}\|_1 + \epsilon} + \frac{\|X_{z=-1} - X_{z=+1}\|_1}{\|X_{z=-1}\|_1 + \|X_{z=+1}\|_1 + \epsilon} \right), \quad (7)$$

so higher values indicate stronger periodic boundary agreement.

3.4.3. Surrogate predictor and validation. We train a 3D CNN surrogate $P(\cdot)$ to predict $\hat{\mathbf{p}}(X)$. When homogenization labels are available on a subset, we validate the surrogate on a held-out labeled set and report predictive error; at test time we use the surrogate for fast verification.

3.4.4. Constraint checking and ranking. A candidate is feasible if it satisfies constraints $\mathbf{c}$ (density bounds, minimum periodicity, *etc.*). We rank feasible candidates by target matching:

$$s(X) = -\|\hat{\mathbf{p}}_{\mathcal{T}}(X) - \mathbf{t}\|_2 - \lambda \text{penalty}(X; \mathbf{c}), \quad (8)$$

and return the top- k .

4. EXPERIMENTS

We evaluate four practical questions: (i) target matching, (ii) constraint satisfaction, (iii) multi-modality, and (iv) interactive efficiency.

4.1. Experimental Setup

4.1.1. Data. We use OpenSS-SDF with $N = 20,000$ samples (train/val/test = 70/15/15), grid 32^3 , domain $[-1, 1]^3$.

4.1.2. Homogenization labels and surrogate verification. To obtain a high-fidelity supervision signal for stiffness in a controlled setting, we select a stratified subset from the training split, compute voxel-based periodic homogenization, and record the resulting effective modulus as E_x^{true} . We then train a 3D CNN surrogate to predict E_x from SDF volumes; at evaluation time, we use E_x^{pred} for fast verification and ranking in experiments.

4.1.3. Prompt suite. We evaluate a template-based prompt suite derived from the held-out test split. Each prompt maps to a structured request $\mathcal{R} = \{\mathbf{t}, \mathbf{c}, k\}$, where $\mathbf{t}$ includes a stiffness target in x , $\mathbf{c}$ includes density and periodicity constraints when specified, and k is the requested number of candidates. In the present suite we use $N_q = 200$ prompts with $k = 5$ candidates per prompt, and we report strict-tolerance success using a relative tolerance on E_x of $\tau = 0.1$.

4.1.4. Implementation details. All models and data processing are implemented in PyTorch. For verification, we use a labeled subset of the training split obtained by voxel-based periodic homogenization (Appendix A.1). We train a lightweight 3D CNN surrogate to predict E_x from the SDF volume and use its prediction E_x^{pred} for fast filtering and ranking at test time. For retrieval, property vectors are precomputed once per split and cached to disk; evaluation then reduces to constraint filtering, nearest-neighbor search, and diversity selection.

4.2. Baselines

In this initial study, we focus on a strong **retrieval baseline** that serves as both a practical method and a sanity check for the evaluation protocol:

- **Retrieval (NN):** nearest-neighbor retrieval in verified property space with constraint filtering and diversity selection.

We leave parametric family fitting and learned generative baselines to follow-on work, and discuss these directions in Section 5.

4.3. Metrics

For each prompt j , the system returns k candidates with verified properties $\hat{\mathbf{p}}^{(i)}$.

4.3.1. Target-matching error.

$$\text{Err}(X) = \|\hat{\mathbf{p}}_{\mathcal{T}}(X) - \mathbf{t}\|_2. \quad (9)$$

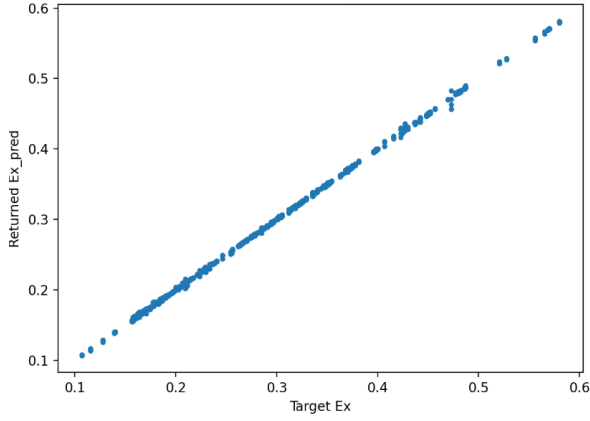


FIGURE 1: Global quantitative results for the retrieval-based conversational pipeline on the held-out prompt suite. Heatmaps visualize alignment between requested targets and the verified properties of retrieved candidates; the bar chart reports strict-tolerance Success@ τ computed from relative error in E_x .

4.3.2. Strict-tolerance success rate. To provide an interpretable strict criterion for stiffness target matching, we report Success@ τ based on relative error in E_x :

$$\text{Success@}\tau = \frac{1}{N_q} \sum_{j=1}^{N_q} \mathbb{I}(\exists i : \text{feas}(X^{(i)}; \mathbf{c}_j) = 1, \frac{|E_x(X^{(i)}) - t_{E_x}|}{|t_{E_x}| + \epsilon} \leq \tau), \quad (10)$$

where ϵ is a small constant for numerical stability.

4.3.3. Feasibility rate.

$$\text{Feasible} = \frac{1}{N_q k} \sum_{j=1}^{N_q} \sum_{i=1}^k \mathbb{I}(\text{feas}(X^{(i)}; \mathbf{c}_j) = 1). \quad (11)$$

4.3.4. Diversity among valid solutions. We compute average pairwise SDF distance among feasible candidates for the same prompt:

$$\text{Div}(j) = \frac{2}{m(m-1)} \sum_{u < v} \|X^{(u)} - X^{(v)}\|_2, \text{ over feasible } m \leq k. \quad (12)$$

4.3.5. Compute cost. We report mean end-to-end latency per prompt (parse + retrieve + verify + rank).

4.4. Main Quantitative Results

Figure 1 visualizes global performance of the retrieval-based conversational pipeline on held-out prompts. Heatmaps show alignment between requested targets and the verified properties of returned candidates, and the bar chart reports strict-tolerance Success@ τ . On this controlled prompt suite, retrieval achieves high feasibility under the specified density and periodicity constraints, and returns candidates whose stiffness matches the requested targets within $\tau = 0.1$ relative tolerance.

Prompt 0: Design a periodic unit cell with low stiffness, medium density, targets: {'Ex': 0.19798903, 'rho': 0.5809326, 's_per': 0.0015985171}
rank 0: sample_id=11403, Ex=0.1981, rho=0.5811, s_per=0.0013
rank 1: sample_id=1491, Ex=0.1974, rho=0.5801, s_per=0.0014
rank 2: sample_id=8648, Ex=0.1971, rho=0.5812, s_per=0.0012
rank 3: sample_id=9605, Ex=0.1990, rho=0.5806, s_per=0.0012
rank 4: sample_id=3670, Ex=0.1978, rho=0.5821, s_per=0.0012

Prompt 1: Design a periodic unit cell with high stiffness, high density, targets: {'Ex': 0.43723693, 'rho': 0.6642456, 's_per': 0.00094366074}
rank 0: sample_id=9977, Ex=0.4370, rho=0.6631, s_per=0.0006
rank 1: sample_id=8912, Ex=0.4365, rho=0.6630, s_per=0.0008
rank 2: sample_id=6354, Ex=0.4379, rho=0.6667, s_per=0.0006
rank 3: sample_id=2363, Ex=0.4345, rho=0.6655, s_per=0.0009
rank 4: sample_id=2707, Ex=0.4372, rho=0.6674, s_per=0.0009

Prompt 2: Design a periodic unit cell with high stiffness, medium density targets: {'Ex': 0.33636898, 'rho': 0.5735779, 's_per': 0.001922965}
rank 0: sample_id=9654, Ex=0.3368, rho=0.5745, s_per=0.0019
rank 1: sample_id=5917, Ex=0.3358, rho=0.5720, s_per=0.0023
rank 2: sample_id=4257, Ex=0.3349, rho=0.5720, s_per=0.0019
rank 3: sample_id=12575, Ex=0.3345, rho=0.5752, s_per=0.0020
rank 4: sample_id=4637, Ex=0.3337, rho=0.5745, s_per=0.0020

FIGURE 2: Qualitative case studies for representative prompts. For each prompt, we show the top-ranked retrieved candidate under verification and ranking, together with diverse alternatives selected from the retrieved pool.

4.5. Qualitative Case Studies

Figure 2 shows representative prompts and the returned candidates. For each prompt, we display the top-ranked candidate under the verify–rank criterion and several diverse alternatives from the retrieved pool. The examples highlight that multiple distinct unit cells can satisfy the same target within tolerance, emphasizing the multi-modal nature of the inverse problem.

4.6. Ablations and Failure Modes

4.6.1. No verification/filtering. Removing constraint checking increases apparent diversity but significantly reduces feasibility and Success@ τ , confirming verification is essential for engineering reliability.

4.6.2. No conversational front-end. Direct numeric conditioning can work when the user provides clean numeric specifications, but fails to support common qualitative intent and mixed constraints, reducing usability.

4.6.3. Common failure modes. We observe (i) contradictory requests (e.g., extreme stiffness at extremely low density) leading to no feasible candidates, (ii) resolution limits hiding thin features relevant to manufacturability, and (iii) surrogate extrapolation causing mis-ranking; uncertainty-aware verification is a promising extension.

5. CONCLUSION

We presented a conversational inverse design framework for cellular unit cells directly in implicit field space. The system translates natural-language intent into explicit targets and constraints, proposes multiple candidate SDF unit cells, and verifies and ranks candidates using fast property prediction validated against homogenization when available. This verify–rank loop makes the interface practical: users can explore a multi-modal design space while maintaining quantitative performance guarantees.

Future work will focus on higher-resolution implicit representation and selection, uncertainty-aware verification (including

selective homogenization calls), stronger manufacturability constraints (minimum feature size, connectivity, printability), and multi-turn refinement where users iteratively steer trade-offs (e.g., “keep option 2 but make it more isotropic”) while preserving constraints. Extending from unit cells to graded and partially aperiodic structures is another promising direction. Finally, we plan to integrate parametric family fitting and learned generative models as additional candidate proposal modules within the same verification protocol.

APPENDIX A. ADDITIONAL IMPLEMENTATION DETAILS

A.1. Voxel FEM Homogenization (Subset Labels)

We compute high-fidelity stiffness labels on a stratified subset of the training split using a voxel-based linear-elastic homogenization procedure with periodic boundary conditions. Each SDF is converted to a binary occupancy field by thresholding at the zero level set. We solve the periodic cell problem on a regular hexahedral voxel grid (Q1 elements) and obtain the effective stiffness tensor C_{eff} . The directional modulus E_x^{true} is extracted from C_{eff} following standard homogenization conventions. In practice, we use an iterative conjugate-gradient solver with a fixed tolerance and maximum iterations; samples that do not converge within the iteration budget are skipped and replaced to keep the labeled subset size fixed.

A.2. Surrogate Architecture

We use a compact 3D CNN regressor to predict E_x from an input SDF volume $X \in \mathbb{R}^{1 \times 32 \times 32 \times 32}$. The network consists of four 3D convolutional blocks (Conv3D + normalization + non-linearity) with strided downsampling, followed by global average pooling and a two-layer MLP head. Inputs are normalized per-volume to zero mean and unit variance using the summary statistics stored in metadata. We train with Adam, early stopping on validation loss, and report predictive error on a held-out labeled split in terms of MAE and R^2 .

A.3. Prompt Templates

We construct a template-based prompt suite to map natural language intent to the structured request $\mathcal{R} = \{\mathbf{t}, \mathbf{c}, k\}$. Templates combine (i) a stiffness target in x (specified as a percentile of the labeled distribution or as a qualitative descriptor such as “stiffer”/“softer”), (ii) optional density bounds (“lower density” $\rightarrow$ lower quantiles of ρ), and (iii) a minimum periodicity requirement (“more periodic” $\rightarrow$ higher thresholds on s_{per}). Each instantiated prompt is paired with ground-truth numeric targets and constraints derived from dataset statistics, making evaluation reproducible.

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